WORKED SOLUTIONS

Pure Unit 3 – Proof

2.3.1 Proof	
Proof by contradiction (including proof of the irrationality of $\sqrt{2}$ and the infinity of primes, and application to unfamiliar proofs).	

- Proof by contradiction P2
- Infinity of primes P6

Proof By Contradiction

Proof is concerned with the demonstration of the truth of an assertion. The essence of proof by contradiction is that we assume that assertion is false and show that this assumption leads to a contradiction.

Example 1

Prove that if n^2 is even then n is even.

Given that n^2 is even, assume that n is not even, i.e. n is odd.

If n is odd, let $n = 2k+1$ where k is an integer.

$$\Rightarrow n^2 = (2k+1)^2 = 4k^2 + 4k + 1$$
$$= 2(2k^2 + 2) + 1$$
$$= \text{Even} + 1$$
$$= \text{Odd}$$

But we are told n^2 is even $\Rightarrow$ By contradiction, the assumption is false and n is even.

Example 2

Show that $\sqrt[3]{2}$ is irrational, i.e. cannot be expressed in the form $\frac{r}{s}$ where r and s are integers.

Assume that $\sqrt[3]{2} = \frac{r}{s}$, written in simplest form i.e. r and s are integers, and r and s do not share any common factors (if they did have common factors then they'd be cancelled out until the fraction was in simplest form).

$$\sqrt[3]{2} = \frac{r}{s}$$
$$2 = \frac{r^3}{s^3}$$
$$2s^3 = r^3$$

So r is a multiple of 2

Let $r = 2k$

$$\Rightarrow 2s^3 = (2k)^3$$
$$2s^3 = 8k^3$$
$$s^3 = 4k^3$$

So s is a multiple of 2.

So r and s have a common factor of 2 and hence the assumption is false.

$\Rightarrow$ By contradiction, $\sqrt[3]{2}$ is irrational.

Example 3

Given that $f(x)$ is a polynomial of degree n, show that $f(x) = 0$ cannot have more than n distinct roots.

Assume that $f(x) = 0$ has more than n distinct roots.

Then the equation has at least $n+1$ roots.

Let these roots be $a_1, a_2, a_3, \ldots, a_{n+1}$.

Then $(x-a_1), (x-a_2), \ldots, (x-a_{n+1})$ are all factors of $f(x)$.

If $(x-a_1)$ and $(x-a_2)$ are both factors of $f(x)$, but not factors of eachother, then $(x-a_1)(x-a_2)$ is a factor of $f(x)$.

$\Rightarrow (x-a_1)(x-a_2)\ldots(x-a_{n+1})$ is a factor of $f(x)$.

This has degree $n+1$ so cannot divide into $f(x)$, which has degree n.

$\Rightarrow$ By contradiction, $f(x)=0$ cannot have more than n distinct root for $f(x)$ with degree n.

Example 4

Use proof by contradiction to show that if a and b are real numbers, then

$$a^2 + b^2 \geq 2ab$$

Assume that $a^2 + b^2 < 2ab$.

Then $a^2 + b^2 - 2ab < 0$

$(a-b)^2 < 0$

Since the square of a real number cannot be negative, $a-b$ is not a real number.

But we are told a and b are real numbers $\Rightarrow a-b$ is a real number.

$\Rightarrow$ By contradiction, the assumption is false and $a^2+b^2 \geq 2ab$.

Example 5

Use proof by contradiction to show that if $(x+1)(x-3) < 0$ *then* $-1 < x < 3$.

Assume that $(x+1)(x-3) < 0$ and either $x \leq -1$ or $x \geq 3$.

If $x \leq -1$, then $(x+1)(x-3) = (\leq 0) \times (< 0)$

Negative $\times$ Negative ≥ 0 and $0 \times$ Negative $= 0$

$\Longrightarrow$ By contradiction, $x > -1$.

If $x \geq 3$, then $(x+1)(x-3) = (\geq 4) \times (\geq 0)$

Positive $\times$ Positive ≥ 0 and Positive $\times 0 = 0$

$\Longrightarrow$ By contradiction, $x < 3$.

So the assumption is false and if $(x+1)(x-3) < 0$, then $-1 < x < 3$.

Example 6

Use proof by contradiction that if x *is real and* $x > 0$ *then* $x + \frac{4}{x} \geq 4$.

Assume that $x + \frac{4}{x} < 4$.

Then $x^2 + 4 < 4x$

$x^2 - 4x + 4 < 0$

$(x-2)^2 < 0$

But all square numbers are ≥ 0 for real values of x.

$\Longrightarrow$ By contradiction, $x + \frac{4}{x} \geq 4$.

Proof by contradiction may sometimes be used to prove a function is one-to-one, i.e. if $f(a) = f(b)$, then $a = b$.

Example 7

Show that the function $f(x) = x^2 - 2x + 3$ for $x > 1$ is one-to-one.

Assume that $f(x)$ is not a one-to-one function so that there exist a and b such that $f(a) = f(b)$ and $a \neq b$.

If $f(a) = f(b)$,

$$a^2 - 2a + \cancel{3} = b^2 - 2b + \cancel{3}$$

$$a^2 - 2a = b^2 - 2b$$

$$a^2 - b^2 - 2a + 2b = 0$$

$$\underline{(a-b)}(a+b) - 2\underline{(a-b)} = 0$$

Common factor.

$$(a-b)(a+b-2) = 0$$

If $a - b = 0$

$a = b$ $\Rightarrow$ False because we assumed $a \neq b$.

If $a + b - 2 = 0$

$a + b = 2$ $\Rightarrow$ False because if $x > 1$ then $a > 1$ and $b > 1$, so $a + b > 2$.

$\Rightarrow$ By contradiction, $f(x) = x^2 - 2x + 3$ for $x > 1$ is one-to-one.

The Infinity of Primes – Proof by Contradiction

This is a 'classic' example of a statement that can be proven easily using proof by contradiction.

There are infinite prime numbers.

Assume that there exists is a **finite** list of **all** the prime numbers. We put them in order from smallest to largest, so call the first one p_1, then next p_2, and so on, until we get to the end of the list, which we can call p_n (the largest prime number).

So our list of all prime numbers looks like $p_1,\ p_2,\ p_3, \dots, p_n$.

Consider the number $Q = p_1 p_2 p_3 \dots p_n + 1$, i.e. the product of all the primes in the list $+1$.

Now one of two scenarios could arise:

1) Q is a prime number:
   If Q is a prime number, it is clearly larger than all the primes in our previous list, which contradicts the fact we had a list of all prime numbers (as Q is not on it) and Q is certainly larger than p_n which was the final number in our list. This contradicts the fact we could write a finite list of all prime numbers.

2) Q is not a prime number:
   We know that all composite (non-prime) numbers can be written as a product of prime factors, so every composite number must be divisible by at least one prime number.
   $Q = p_1 p_2 p_3 \dots p_n + 1$ cannot be divided by p_1, as it would leave a remainder of 1, and cannot be divided p_2 or p_3 or any of the primes in the list. So Q is not a prime number but cannot be divided by any of the primes in the list. So, there must exist another prime number which is not on the list, so this contradicts the fact we could write a finite list of all prime numbers.

Therefore, we cannot write a finite list of all prime numbers – there must be an infinite number of prime numbers.

Exercise

Use proof by contradiction to prove the following:

1) Show that if n^3 is odd, then n is odd.

2) Show that if $2n^2 + n$ is even, then n is even.

3) Show that $\sqrt{3}$ is irrational.

4) Show that if x is real and $x > 0$, then $x + \frac{1}{x} \geq 2$.

5) Show that if $(x-2)(x-5) \geq 0$, then $x \leq 2$ or $x \geq 5$.

6) Show that if x and y are real, then $x^2 + 4y^2 \geq 4xy$.

7) Show that if $f(x)$ is a cubic, then $f(x) = 0$ cannot have more than 3 distinct roots.

8) Show that the function defined by $f(x) = x^3 + 4x + 5$ for $x \leq -2$ is one-to-one.

9) Show that the function defined by $f(x) = x^2 + 6x + 9$ for $x \geq 0$ is one-to-one.

10) [WJEC 2018 Q11] For every real number x such that $0 \leq x \leq \frac{\pi}{2}$, $\sin x + \cos x \geq 1$.

Solutions:

A2 - PROOF - EXERCISE FROM NOTES - SOLUTIONS

PROOF BY CONTRADICTION.

① SHOW THAT IF n^3 IS ODD, THEN 'n' IS ODD.

ASSUME 'n' IS EVEN $\therefore n = 2k$

$$n^3 = (2k)^3$$
$$n^3 = 8k^3$$
$$n^3 = 2(4k^3)$$

$\therefore n^3$ IS DIVISIBLE BY 2, ie IT IS EVEN,
BUT n^3 IS ODD, THEREFORE A CONTRADICTION.
THE ASSUMPTION IS FALSE AND 'n' IS ODD.

② SHOW THAT IF $2n^2+n$ IS EVEN, THEN 'n' IS EVEN.

ASSUME 'n' IS ODD $\therefore n = 2k+1$

$$\therefore 2n^2+n = 2(2k+1)^2 + (2k+1)$$
$$= 2[4k^2+4k+1] + 2k+1$$
$$= 8k^2+8k+2+2k+1$$
$$= 8k^2+10k+2+1$$
$$= 2[4k^2+5k+1]+1$$

WHICH IS ODD, BUT $2n^2+n$ IS EVEN,
THEREFORE A CONTRADICTION.
THE ASSUMTION IS FALSE AND 'n' IS EVEN.

③ SHOW THAT $\sqrt{3}$ IS IRRATIONAL. (ie CANNOT BE WRITTEN AS $\frac{a}{b}$)

ASSUME $\sqrt{3}$ IS RATIONAL $\therefore$ CAN BE WRITTEN AS $\frac{a}{b}$ WHERE 'a' AND 'b' ARE INTEGERS HAVING NO COMMON FACTORS.

$$\sqrt{3} = \frac{a}{b}$$

$$(\sqrt{3})^2 = \left(\frac{a}{b}\right)^2$$

$$3 = \frac{a^2}{b^2} \longrightarrow 3b^2 = a^2$$

$\therefore \quad 3b^2 = a^2$

$\therefore$ a^2 IS DIVISIBLE BY 3.

$\therefore$ a IS DIVISIBLE BY 3, $a = 3m$

SUB $a = 3m$ INTO

$$3b^2 = a^2$$
$$3b^2 = (3m)^2$$
$$3b^2 = 9m^2$$
$$b^2 = 3m^2$$

$\therefore$ b^2 IS DIVISIBLE BY 3.

$\therefore$ b IS DIVISIBLE BY 3, $b = 3n$

$\therefore$ 'a' AND 'b' HAVE A COMMON FACTOR, BUT 'a' AND 'b' HAVE NO COMMON FACTORS THEREFORE A CONTRADICTION.

THE ASSUMPTION IS FALSE AND $\sqrt{3}$ IS IRRATIONAL.

(4) SHOW THAT IF 'x' IS REAL AND $x > 0$, THEN

$$x + \frac{1}{x} \geq 2$$

ASSUME $x + \frac{1}{x} < 2$ $(\times x)$

$$x^2 + 1 < 2x$$
$$x^2 - 2x + 1 < 0$$

[PERFECT SQUARE] $(x-1)^2 < 0$

BUT $(x-1)^2$ MUST BE GREATER THAN OR EQUAL TO ZERO IF 'x' IS REAL.

(THE SQUARE OF A REAL NUMBER CANNOT BE NEGATIVE) $\therefore$ CONTRADICTION

THE ASSUMPTION IS FALSE AND $x + \frac{1}{x} \geq 2$

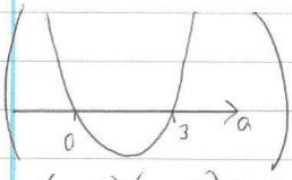
(5) SHOW THAT IF $(x-2)(x-5) \geq 0$, THEN $x \leq 2$ OR $x \geq 5$.

ASSUME $x > 2$ OR $x < 5$

$\underline{x > 2}$ $\quad x = 2 + a$ WHERE $a > 0$

$\therefore (x-2)(x-5) = (2+a-2)(2+a-5)$

$= a(a-3)$



$a(a-3) < 0$ FOR $0 < a < 3$

$\therefore (x-2)(x-5) < 0$, BUT

$(x-2)(x-5) \geq 0$, THEREFORE A CONTRADICTION.

<u>THE ASSUMPTION IS FALSE AND $x \leq 2$.</u>

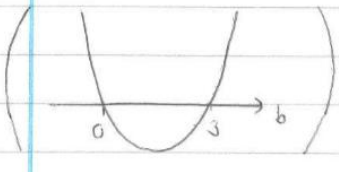
$\underline{x < 5}$ $\quad x = 5 - b$ WHERE $b > 0$

$\therefore (x-2)(x-5) = (5-b-2)(5-b-5)$

$= (3-b)(-b)$

$= -3b + b^2$

$= b(b-3)$



$b(b-3) < 0$ FOR $0 < b < 3$,

$\therefore (x-2)(x-5) < 0$, BUT $(x-2)(x-5) \geq 0$, THEREFORE A CONTRADICTION.

<u>THE ASSUMTION IS FALSE AND $x \geq 5$.</u>

(6) SHOW THAT IF 'x' AND 'y' ARE REAL, THEN

$$x^2 + 4y^2 \geq 4xy$$

ASSUME THAT, IF 'x' AND 'y' ARE REAL, THEN $x^2 + 4y^2 < 4xy$

$\therefore x^2 - 4xy + 4y^2 < 0$

[PERFECT SQUARE] $(x-2y)^2 < 0$

BUT $(x-2y)^2$ MUST BE GREATER THAN OR EQUAL TO ZERO IF 'x' AND 'y' ARE REAL.

$\therefore$ CONTRADICTION.

<u>THE ASSUMPTION IS FALSE AND $x^2 + 4y^2 \geq 4xy$</u>

(7) SHOW THAT IF $f(x)$ IS A CUBIC, THEN $f(x) = 0$ CANNOT HAVE MORE THAN 3 DISTINCT ROOTS

ASSUME $f(x)$ HAS MORE THAN 3 DISTINCT ROOTS.
THEN THE EQUATION HAS AT LEAST 4 ROOTS, a_1, a_2, a_3, a_4.

$$\therefore f(x) = (x-a_1)(x-a_2)(x-a_3)(x-a_4)$$
$$(x-a_1)(x-a_2)(x-a_3)(x-a_4) = 0$$
$$x^4 + \cdots = 0$$

THIS IS A POLYNOMIAL OF DEGREE 4, BUT $f(x)$ IS A POLYNOMIAL OF DEGREE 3 (A CUBIC), THEREFORE A CONTRADICTION.

<u>THE ASSUMPTION IS FALSE AND $f(x) = 0$ CANNOT HAVE MORE THAN 3 DISTINCT ROOTS.</u>

(8) SHOW THAT THE FUNCTION DEFINED BY $f(x) = x^2 + 4x + 5$ FOR $x \leq -2$ IS ONE-ONE.

ASSUME $f(x)$ IS NOT ONE-ONE FOR $x \leq -2$.
SO THERE EXISTS 'a' AND 'b' SUCH THAT $f(a) = f(b)$ AND $a \neq b$.

THEN
$$a^2 + 4a + 5 = b^2 + 4b + 5$$
$$a^2 - b^2 + 4a - 4b = 0$$
$$(a-b)(a+b) + 4(a-b) = 0$$
$$(a-b)(a+b+4) = 0$$

$a \neq b \therefore (a-b) \neq 0$, DIVIDING THROUGHOUT BY $(a-b)$ GIVES $a+b+4 = 0$

NOW $x \leq -2$ SO $a \leq -2$ AND $b \leq -2$
GIVING $a+b \leq -4$, THAT IS $a+b+4 \leq 0$,
THEREFORE A CONTRADICTION.

<u>THE ASSUMPTION IS FALSE AND $f(x)$ IS ONE-ONE.</u>

(9) SHOW THAT THE FUNCTION DEFINED BY

$$f(x) = x^2 + 6x + 9 \quad \text{FOR} \quad x \geq 0 \quad \text{IS ONE-ONE.}$$

ASSUME $f(x)$ IS NOT ONE-ONE FOR $x \geq 0$

SO THERE EXISTS 'a' AND 'b' SUCH THAT

$f(a) = f(b)$ AND $a \neq b$.

THEN $a^2 + 6a + 9 = b^2 + 6b + 9$

$$a^2 - b^2 + 6a - 6b = 0$$

$$(a-b)(a+b) + 6(a-b) = 0$$

$$(a-b)(a+b+6) = 0$$

$a \neq b \therefore (a-b) \neq 0$, DIVIDING THROUGHOUT BY $(a-b)$ GIVES $a+b+6=0$.

NOW $x \geq 0$ SO $a \geq 0$ AND $b \geq 0$

GIVING $a+b \geq 0$, SO $a+b+6 \geq 6$,

THEREFORE A CONTRADICTION.

<u>THE ASSUMPTION IS FALSE AND $f(x)$ IS ONE-ONE.</u>

10) [WJEC 2018 Q11] For every real number x such that $0 \leq x \leq \frac{\pi}{2}$, $\sin x + \cos x \geq 1$.

Assume $\sin x + \cos x < 1$. We know $\sin x$ and $\cos x$ are both non-negative for $0 \leq x \leq \frac{\pi}{2}$, so we have

$$0 < \sin x + \cos x < 1$$

Hence if we square both sides of our assumption:

$$\sin x + \cos x < 1$$
$$(\sin x + \cos x)^2 < 1^2$$
$$\sin^2 x + 2\sin x \cos x + \cos^2 x < 1$$
$$1 + \sin 2x < 1$$
$$\sin 2x < 0$$

But, for $0 \leq x \leq \frac{\pi}{2}$, $\sin 2x \geq 0$ as $\sin x \geq 0$ for $0 \leq x \leq \pi$.
so this is a contradiction, hence $\sin x + \cos x \geq 1$ //

Mark Scheme:

Suppose that $\sin x + \cos x \geq 1$ is not true. Then there exists an x in the given domain		
for which $\sin x + \cos x < 1$	M1	
$(\sin x + \cos x)^2 < 1^2$	A1	
$\sin^2 x + 2\sin x\cos x + \cos^2 x < 1^2$		
$1 + 2\sin x\cos x < 1$		
$\sin x\cos x < 0$	A1	or $\sin 2x < 0$
As $\sin x \geq 0$ and $\cos x \geq 0$, this is impossible.		
Hence $\sin x + \cos x < 1$ cannot be true,		
hence $\sin x + \cos x \geq 1$ for $0 \leq x \leq \frac{\pi}{2}$.	A1	cso